Mathematics General

HSC Marking Feedback 2018

Question 26

Students should:

- be familiar with the terminology used in all topics
- always show substitution into any given formula
- consider the size practicality of any answers they provide
- take a ruler into the examination.

In better responses, students:

- presented more than one ‘guess and check’ possibility (part b)
- understood the term ‘relative frequency’ (part a)
- used clear and concise explanations of standard deviation or spread of data.
- correctly used ratios or fractions to show a comparison (part g i)
- were able to calculate the required areas and find the quantity of fertiliser by multiplication by 26.5.

Areas for students to improve include:

- identifying the difference between frequency and cumulative frequency
- entering data into their calculator
- converting lengths, using their scale, before finding the area
- differentiating between the terms interquartile range and range (part e).

Question 27

Students should:

- always refer to the Reference Sheet to ensure the correct version of the required formula is used

- present full working out for solutions worth multiple marks.

In better responses, students:

- solved simultaneous equations using the elimination method (part b)
- converted cents to dollars at the start of a calculation (part a)
- correctly interpreted information from a graph (part d).

Areas for students to improve include:

- calculating the radius, given the diameter
- calculating the curved surface area of a cylinder
- writing a formal equation given the gradient and y-intercept
- reviewing the concept of z-scores.

Question 28

Students should:

- clearly link their steps of working together
- apply the correct number of applications required for the trapezoidal rule
- familiarise themselves with fraction usage in equations.

In better responses, students:

- substituted correctly into a given formula (part a)
- used percentages correctly (parts c and d i)
- converted a yearly interest rate to a daily rate (part d i).

Areas for students to improve include:

- converting kilometres to metres, hours to seconds, watts to kilowatts (part c and e)
- calculating the reaction-time distance
- understanding how to answer a 'show' question.

Question 29

Students should:

- underline or highlight important information in a question
- understand the context of the equation of a straight line when it is presented in the form $y = mx + b$.

In better responses, students:

- calculated correct flying times considering time differences
- correctly read data from a table and substituted the correct data into a formula
- were able to calculate savings made by paying off a loan earlier.

Areas for students to improve include:

- converting from MB to bits and bytes
- working with inverse variation
- providing explanations for a situation using both calculations and comments supporting the calculations (part d ii).

Question 30

Students should:

- understand the difference between cubic metres and litres
- carefully read whether surface area of volume is required
- use the number of lines as an indication or the expected length of the response.

In better responses, students:

- used the correct tax bracket and correct taxable income from the information given
- calculated the length AC using the correct trigonometric ratio
- could substitute the length AC, 30 cm^2 and $40°$ into the correct formula
- could correctly state the probabilities on each branch and calculate a financial expectation that included the subtraction of the game cost.

Areas for students to improve include:

- calculating tax payable from a table and understanding the difference between net income and taxable income (part b)
- applying the correct area formula for right and non-right angled triangles
- calculating probabilities that involve multiplying along the branches of correctly completed probability trees (part d).